

Lecture 4 : Latent space and Guidance

- high dimensionality $\longrightarrow$ tractable dimension
- redundant information (correlated Pixels) $\longrightarrow$ compact representation
- semantically not meaningful space $\longrightarrow$ meaningful semantic space

semantic similarity & perceptual similarity

Attempt 1: Learn representation via an autoencoder.

Spatial compression ratio: $f = \frac{H}{h} = \frac{W}{w}$

Encoder: convolution + pooling

high dim $\longrightarrow$ low dim

Decoder: low dim $\longrightarrow$ high dim z

Loss
$$\mathcal{L}_{AE} = \| x - \underbrace{D_{\theta}(E_{\varphi}(x))}_{\hat{x}} \|^2$$

$$q_{\varphi}(\cdot|x) = \mathcal{N}(\mu_{\varphi}(x), \Sigma_{\varphi}^2(x))$$

$$q(z|x) \approx p(z).$$

$\longrightarrow$ this is approximately a posterior distribution

$\max_{\theta} \log p_{\theta}(x)$ objective

Variational Autoencoder

$$p_{\theta}(x) = \int p_{\theta}(x, z) dz$$

$$p(z) p_{\theta}(x|z)$$

output of the encoder

$$p_{\theta}(x) = \int \frac{p(z) p_{\theta}(x|z)}{p_{\phi}(z|x)} p_{\phi}(z|x) dz$$

output of the decoder

$$= \mathbb{E}_{z \sim p_{\phi}(z|x)} \left[\frac{p(z) p_{\theta}(x|z)}{p_{\phi}(z|x)} \right]$$

Jensen's Inequality $\rightarrow$ lower bound.

$$\log(p_{\theta}(x)) = \log \left(\mathbb{E}_{z \sim p_{\phi}(z|x)} \left[\frac{p(z) p_{\theta}(x|z)}{p_{\phi}(z|x)} \right] \right) \geq \mathbb{E}[\log(\cdot)]$$

show lower bound is tractable...

$$\mathbb{E}[\log(\cdot)] = \mathbb{E}_{z \sim p_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \underbrace{\mathbb{E}_{z \sim p_{\phi}(z|x)} \left[\log \left[\frac{q_{\phi}(z|x)}{p(z)} \right] \right]}_{KL[q_{\phi}(z|x) \| p(z)]}$$

Variational Autoencoder ELBO,

① Derive lower bound for max log likelihood estimation

$$\log p_{\theta}(x) \geq \mathbb{E}_{z \sim q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{q_{\phi}(z|x)} \right]$$

$$\textcircled{2} \quad \text{ELBO} = \underbrace{\mathbb{E}_z [\log p_{\theta}(x|z)]}_{\text{reconstruction}} - \underbrace{KL(q_{\phi}(z|x) \| p(z))}_{\text{regularization of latent space}}$$

• blurry image.

Why do VAEs produce blurry images?

$$\mathcal{L}_{\text{rec}} = \|x - \hat{x}\|^2$$

idea . check reconstruction of input

description . $\mathcal{L}_1$ and $\mathcal{L}_2$ pixel wise distance between output and actual

weight $\lambda_{\text{rec}} \approx 1$. high $\lambda_{\text{rec}} \rightarrow$ blurry output

$$\mathcal{L}_{\text{KL}} = \text{KL}(q_{\phi}(z|x) \| p(z))$$

idea make latent space more structured

description . KL divergence between output of encoder and prior.

weight . $\lambda_{\text{KL}} \approx 10^{-6}$ λ_{KL} high $\rightarrow$ posterior collapse

To combat blurriness: Learned Perceptual Image Patch Similarity (LPIPS).

Make output more realistic with adversarial loss.

Adversarial Loss & GANs.

Diffusion in the latent space: Encode + Diffuse + Decode.

DINO: self distillation with no labels.

Vision Transformer

Contrastive Learning

Guidance

$$p(x_t | x_{t+1}) \longrightarrow p(x_t | x_{t+1}, y)$$

$$p_{\theta, \phi}(x_t | x_{t+1}, y)$$

└── classifier weights

└── generation weights.

• We want to compute $p_{\theta}(x_t | x_{t+1}, y)$

• Bayes Rule: $P(A|B) = P(A \cap B) / P(B) = P(B|A)P(A) / P(B)$

$$P(A|B, C) = P(B|A, C) P(A|C) / P(B|C)$$

$$\text{let } A = x_t, B = y, C = x_{t+1}$$

$$p(x_t | y, x_{t+1}) = p(y | x_t, x_{t+1}) p(x_t | x_{t+1}) / p(y | x_{t+1})$$

markov assumption

$$= p_{\phi}(y | x_t) p_{\theta}(x_t | x_{t+1}) / p(y | x_{t+1})$$

denominator does not
depend on x_t

$$\propto p_{\phi}(y | x_t) p_{\theta}(x_t | x_{t+1})$$

└── classifier

└── unconditional reverse

Detail 1: $p_{\theta}(x_t | x_{t+1}) \sim N(\mu_{\theta}, \sigma_{t+1}^2 I)$

Detail 2: $p_{\phi}(y | x_t) \rightarrow \log p_{\phi}(y | x_t) \approx (x_t - \mu)^T \nabla_{x_t} \log p_{\phi}(y | \mu) + \text{constant}$

$$x_t \approx N\left(\mu_{\theta} + \underbrace{\sigma_{t+1}^2 \nabla_{x_t} \log p_{\phi}(y | \mu_{\theta})}_{\text{shifted mean}}, \sigma_{t+1}^2 I\right)$$

Training

generation model $\rightarrow$ not retrained $\rightarrow$ purely sampling

classifier trained with cross-entropy loss on noisy images

$\hookrightarrow$ sample $x_t \rightarrow$ predict correct label $p_\theta(y|x_t)$.

- give more weight to classifier part $\rightarrow$ more guidance.

Limitations:

new requirements

- classifier train on labeled data
- train on noisy data
- align distribution p data between two models

computational considerations

- extra backwards pass needed at each step
- gradient needs tuning
- approximation via Taylor exp.

implicit guidance $\rightarrow$

classifier based

$$\epsilon_{\theta}(x_t) - w \nabla_{x_t} \log p_{\phi}(y|x_t)$$

classifier free

$$\epsilon_{\theta}(x_t, \emptyset) + w \cdot (\epsilon_{\theta}(x_t, y) - \epsilon_{\theta}(x_t, \emptyset))$$

p_{uncond}

p_{uncond}

